NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY
Faculty of Computing and Informatics
Department of Software Engineering

SDN621S — ASSESSMENT 2

STEMMate Namibia

Individual Requirements-to-Design Portfolio

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This portfolio transforms the refined A1 requirements for the fictional offline-first STEMMate Namibia planning platform into coherent structural, behavioural and interaction designs that can guide a focused implementation.

1. Refined Requirements Baseline

The following 11 prioritised requirements form the design baseline. They incorporate the A2 MVP (browse/filter, offline save, session planning, synchronisation) and explicitly address accessibility, privacy and security. IDs are carried forward from A1 where stable; new or refined items are marked. All requirements are testable.

ID	Requirement	Type	Priority
FR-01	The system shall allow users to browse and filter activities by level, duration, topic and material availability (local cache first).	Functional	Must
FR-02	The system shall provide a “Save for Offline” function that caches activity text (and images when connectivity permits) so that the activity remains usable with zero connectivity.	Functional	Must
FR-03	The system shall allow facilitators to create/edit session plans containing ordered steps, timing, materials list and mandatory safety notes; finalisation is blocked if safety notes are empty.	Functional	Must
FR-04	The system shall present inclusion prompts during planning so that every plan records at least one adaptation consideration.	Functional	Must
FR-05	The system shall allow recording of aggregate participant counts by broad category only (no names, photos or school IDs).	Functional	Must
FR-07	The system shall detect connectivity and synchronise locally saved plans and kit status; conflicts are surfaced and resolved via an explicit strategy without data loss.	Functional	Must
QR-01	On devices with ≤2 GB RAM the main screens shall become interactive within 5 s after launch from a cold start (measured on a representative low-spec Android browser).	Quality	Should
QR-02	The system shall use a local-first store so that an abrupt signal drop or process kill never loses an in-progress draft (zero data-loss under crash).	Quality	Must
QR-03	The UI shall maintain a minimum contrast ratio of 4.5:1 for text and support full keyboard-only navigation of all planning flows (accessibility).	Quality / Accessibility	Must
PR-01	The system shall hard-reject any attempt to enter pupil names, photos or school identifiers; only integer aggregate counts are accepted.	Privacy	Must
SEC-01	A prominent Secure Sign-Out control shall be available on every main screen and shall immediately clear local session tokens.	Security	Must
SEC-02	An automated session timeout (default 5 min idle) shall lock the application on shared devices; the timeout is configurable but never disabled.	Security	Should

Note: FR-06 (kit return tracking) and FR-08 (Manager View) remain Should-Have and are supported by the same SyncService and OfflineStore abstractions but are not required for the minimum viable design modelled below.

2. A1 Feedback and Change Summary

No formal lecturer feedback document was supplied with the A1 artefacts. The changes below are therefore self-identified refinements driven by (a) alignment with the published A2 MVP, (b) improved

testability of acceptance indicators, and (c) explicit treatment of accessibility and shared-device security that were only partially specified in A1. Each change is transparent and does not invent new evidence.

Req. ID	A1 issue / observation	Change	Reason	Effect on design
FR-02	“Save for Offline” stated but partial-download behaviour undefined.	Clarified that textual content is always cached; images best-effort.	Supports intermittent connectivity (E5).	OfflineStore.saveActivity accepts partial payloads; sequence A shows exception path.
FR-03	Safety-note mandatory rule present but not linked to finalisation guard.	Explicit finalisation guard added.	Directly testable acceptance indicator from A1.	SessionPlanService.finalise() raises if safetyNotes empty.
QR-03	High-contrast & keyboard support listed only as operating constraint.	Elevated to Must-Have quality/accessibility requirement.	A2 requires explicit accessibility requirements.	Influences UI component choices; keyboard navigation paths modelled in interaction flow.
SEC-02	Timeout labelled ASSUMPTION in A1.	Retained as Should-Have with clear default (5 min) and non-disableable policy.	Shared-device risk (E5) remains critical.	UserSession.checkTimeout() called on every foreground resume.
FR-07	Sync described; conflict resolution left open.	Conflict resolution made explicit via Strategy pattern.	Required for coherent behavioural model.	ConflictResolver class + sequence B alternative path.
New (none)	No formal interface contract in A1.	IOfflineStorage and ISyncGateway contracts added.	A2 portfolio item 6.	Enables independent testing and future replacement of storage engine.

3. Domain Model

The domain model captures the core STEMMate concepts needed by the MVP. Definitions are supplied for any term whose everyday meaning could be ambiguous.

Activity — a reusable STEM learning unit described by level, duration, topic and a list of required materials.

Material — an inexpensive or locally obtainable item (e.g. cardboard, batteries) used as a filter criterion.

SessionPlan — a facilitator-authored instance of one or more activities for a concrete session, containing ordered steps, timing, safety notes and inclusion prompts.

Kit — a shared physical equipment package whose reservation status must be tracked across offline users.

SyncRecord — metadata that records the last-known version and conflict state of any locally modified entity. **User** — an authenticated adult facilitator or coordinator; pupils are deliberately outside the model.

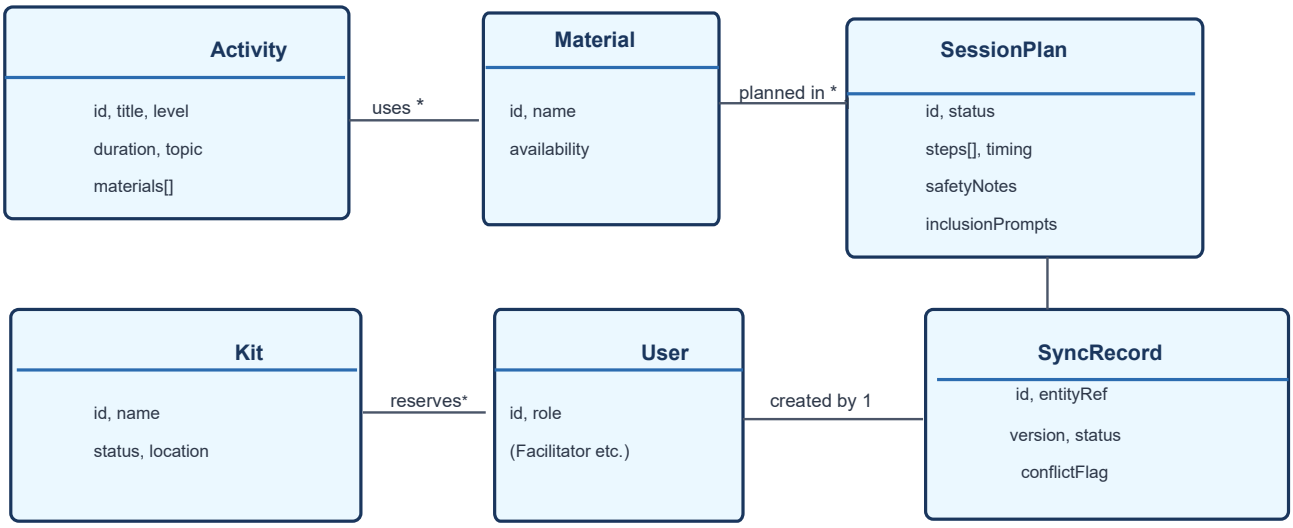


Figure 1 — Domain model (concepts, relationships and multiplicities)

4. Design Class Diagram

Responsibilities are allocated to cohesive services. Persistence and synchronisation are accessed only through narrow interfaces (IOfflineStorage, ISyncGateway) so that the concrete storage engine can be replaced without touching business logic. Visibility is public for service operations that form the application API; internal helpers remain package-private (not shown for brevity).

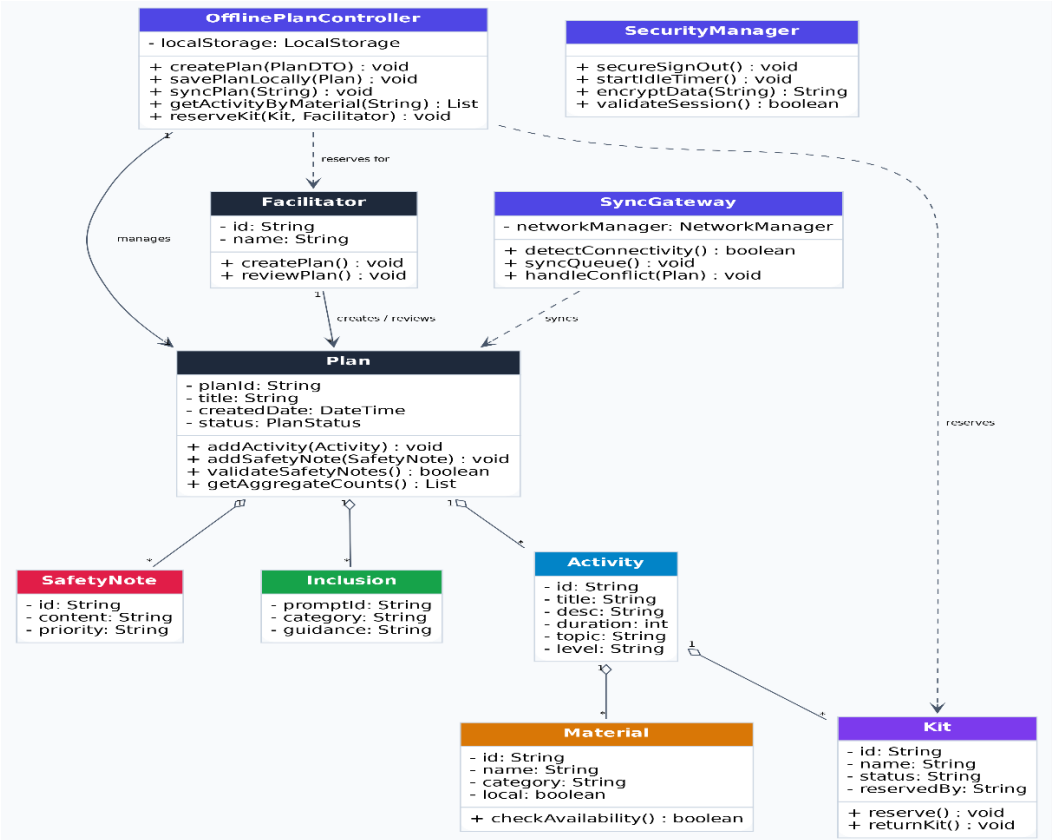


Figure 2 — Design class diagram (services, interfaces, Strategy collaborator)

5. Sequence Diagrams

5.1 Scenario A — Find and save offline (with incomplete download)

A facilitator filters activities by locally available materials, selects one, and requests an offline save. When the network fails mid-download the system still persists the textual description so the activity remains usable (FR-02, QR-02).

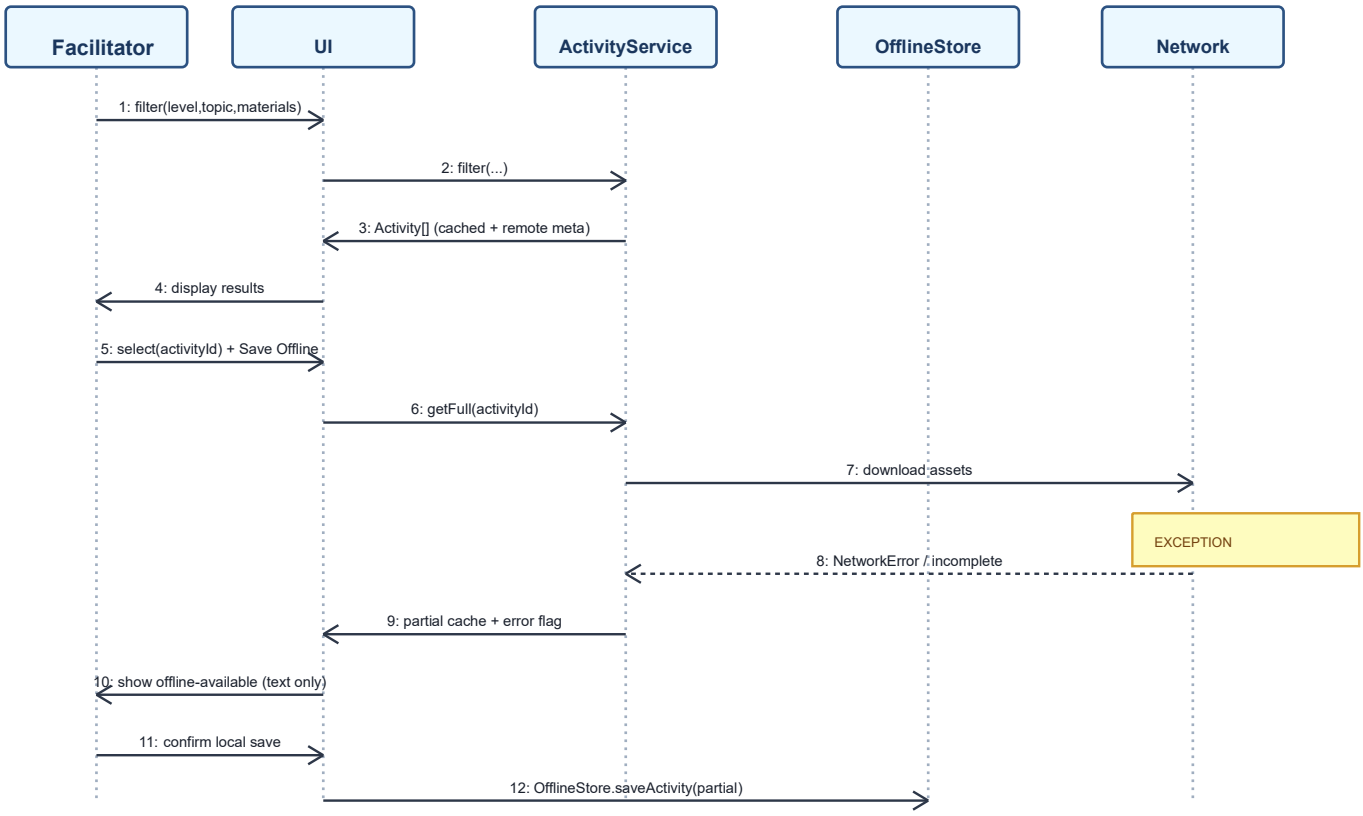


Figure 3 — Sequence A: Find & save offline (exception path shown)

5.1 Scenario B — Plan offline and synchronise (with version conflict)

A facilitator creates or edits a session plan while offline. The draft is written to the local-first store. When connectivity returns the SyncService attempts a push; a remote newer version triggers the ConflictResolver Strategy. The draft is never discarded.

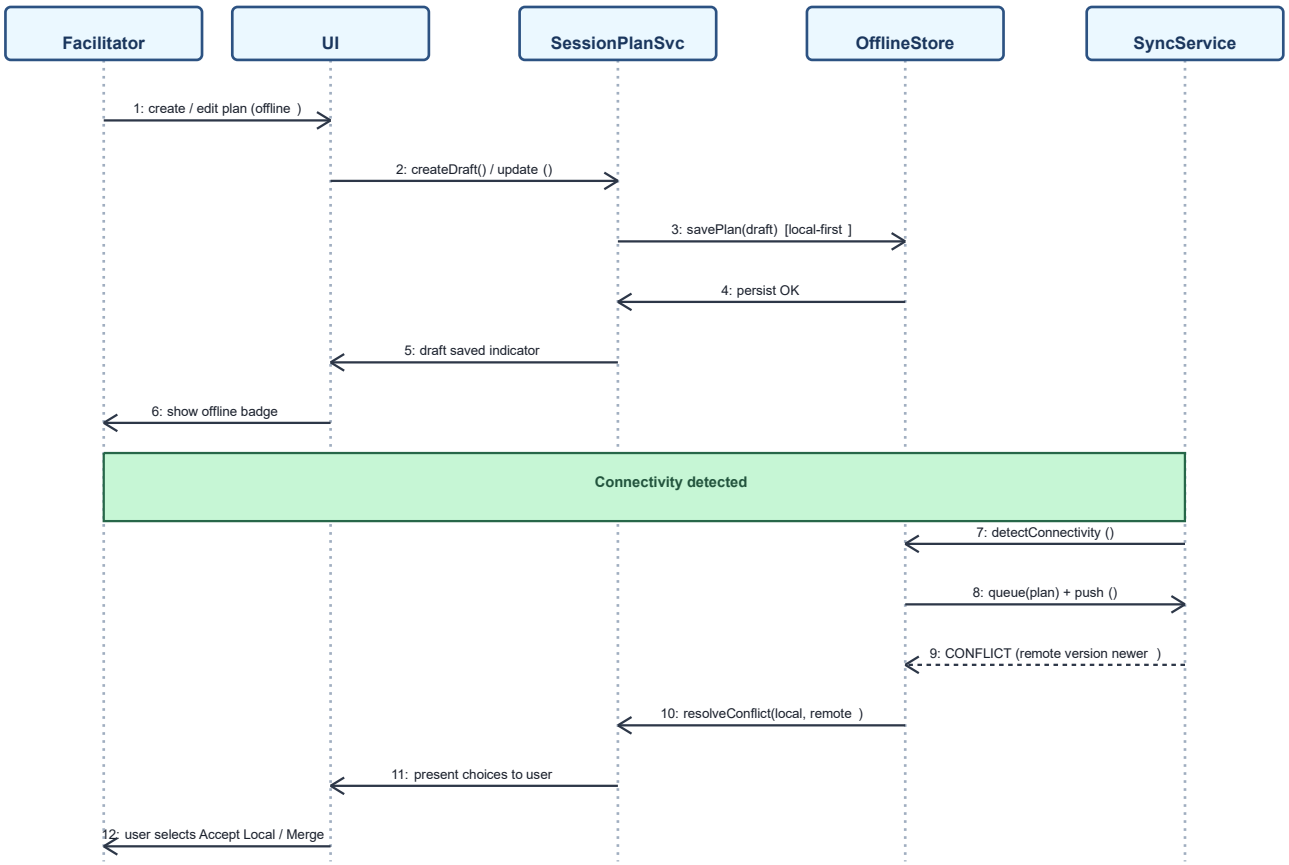


Figure 4 — Sequence B: Plan & synchronise (conflict alternative path)

6. Component / Interface Contract — IOfflineStorage

Purpose: provide a uniform, testable abstraction over the local-first persistence layer so that ActivityService, SessionPlanService and SyncService remain independent of the concrete engine (SQLite.).

Operation	Inputs	Outputs / Effects	Preconditions & Errors
persist(key, data)	key: string data: opaque blob or JSON	void (data durable after return)	Pre: key non-empty. Error: StorageFullError, WriteFailedError
retrieve(key)	key: string	data or null if absent	Error: ReadFailedError (I/O)
exists(key)	key: string	Boolean	Never throws for missing key
remove(key)	key: string	Void	Idempotent; missing key is success
clear()	None	void (all keys removed)	Used only on Secure Sign-Out (SEC-01)

Quality constraints: every write must be atomic with respect to process kill (QR-02); worst-case latency on a 2 GB device for a 50 KB plan payload ≤ 200 ms; no network calls may be made from any implementation of this interface.

7. Design Pattern and Trade-off

Chosen pattern: Strategy (ConflictResolver). Problem addressed: when a local draft and a remote version of the same SessionPlan diverge, the system must apply a resolution policy without hard-coding a single rule inside SyncService. Different deployment contexts may prefer “last-writer-wins”, “facilitator-always-wins”, or “manual merge”. Embedding any one policy would force a code change for each new rule.

In the design, SyncService depends only on the ConflictResolver interface; concrete strategies (AcceptLocal, AcceptRemote, ThreeWayMerge) are injected. Sequence B demonstrates the interaction when a conflict is detected.

Rejected alternative: hard-coded last-writer-wins inside SyncService. Evaluation criteria: (1) adherence to open-closed principle, (2) testability of each resolution path in isolation, (3) ability to evolve policy without touching the synchronisation state machine, (4) cognitive load for future maintainers. Hard-coding scores poorly on (1)–(3). The extra class is justified by the explicit requirement that drafts must never be silently discarded (QR-02) and by the stakeholder tension between Facilitator and Equipment Custodian identified in A1.

8. Interaction Flow — Offline Material Pivot

Low-fidelity flow for the core facilitator task of switching to a locally available activity when expected materials are missing. Offline status is continuously visible; safety-note validation prevents incomplete plans from being marked ready.

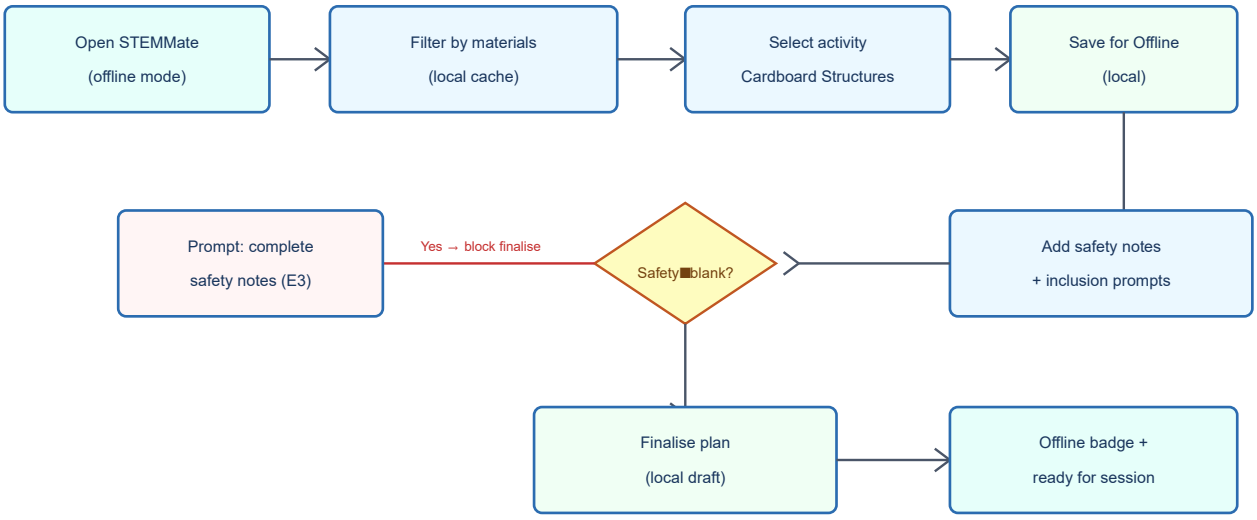


Figure 5 — Interaction flow (offline material pivot + safety guard)

9. Design Traceability and Rationale

9.1 Requirement-to-design traceability

Req. ID	Summary	Model element	Interaction / Contract	Acceptance evidence
FR-01	Filter by materials	ActivityService.filter()	Seq A steps 1–4	Returns relevant results < 2 s offline
FR-02	Offline caching	OfflineStore + IOfflineStorage	Seq A steps 5–12; contract persist()	Textual activity usable after network fail
FR-03	Safety notes mandatory	SessionPlanService.finalise()	Interaction flow diamond	Finalise blocked when safety field null
FR-07	Synchronisation + conflict	SyncService + ConflictResolver	Seq B steps 7–12	Conflict surfaced; draft retained

QR-02	Zero data loss	IOfflineStorage atomic write	Contract quality constraint	Draft survives hard kill
PR-01	No PII	SessionPlan aggregate counts only	Domain model (no Pupil entity)	Text name fields rejected by validation
SEC-01	Secure Sign-Out	UserSession.signOut() + clear()	IOfflineStorage.clear()	Tokens cleared on button press
QR-03	Contrast + keyboard	UI constraints (not structural)	Interaction flow keyboard path	Contrast ≥ 4.5:1; all actions reachable by Tab

9.2 Design rationale

Cohesion & coupling. Each service owns a single responsibility (filtering, planning, synchronisation, security). Coupling is restricted to interface dependencies; concrete storage and network gateways can be replaced independently, improving maintainability and testability (mock IOfflineStorage for unit tests of SessionPlanService).

Maintainability & testability. The Strategy pattern for conflict resolution isolates policy decisions. Acceptance indicators from A1 map directly onto public methods and sequence outcomes, enabling automated checks.

Accessibility, security & privacy trade-offs. Elevating QR-03 to Must-Have forces high-contrast defaults and keyboard reachability, slightly increasing layout complexity but necessary for facilitators with varying digital confidence and possible visual impairment. The non-disableable timeout (SEC-02) trades a minor interruption risk against the far higher risk of residual data on a shared device (E5). Privacy is enforced structurally: the domain model contains no Pupil entity and validation rejects free-text identifiers (PR-01).

Assumptions still open. Typical device RAM and browser versions in the target Namibian context remain unbenchmarked; QR-01 therefore stays Should-Have until field measurements are available. Multilingual readiness is acknowledged but left outside the MVP structural design (icon-first navigation is an interaction assumption only).

References

1. Namibia University of Science and Technology, Department of Software Engineering. (2026). SDN621S Assessment 2: STEMMate NamibiaIndividual requirements-to-design portfolio [Assessment brief]. NUST eLearning.
2. Heita, J. N. (2026). Requirements Analysis Report: STEMMate Namibia (SDN621S Assignment 1).

Artificial Intelligence Declaration

Tool used for language polishing: Claude / DeepSeek (prompts limited to clarity and consistency of English phrasing).

Used Grok for the final diagrams